

APPENDIX B

METRIC CONVERSION CHARTS

B-1 SCOPE.

This appendix contains information for converting values in terms of United States System of Weights and Measures to equivalent IS metric units (International System of Units). This information is not intended to be comprehensive for the classes of measurement conversions provided. Appropriate conversion tables and/or guides, of which there are many, should be consulted for more complete information.

B-2 DESCRIPTION.

This appendix contains conversion tables for converting the more common classes of measurement that may be encountered in this manual: Length, Weight, Temperature, Torque, and Pressure.

B-3 RULES FOR CONVERSION AND ROUNDING.

B-3.1 Quantity.

Conversion of quantities must be handled carefully with regard to the intended accuracy of the data and the given number of digits. In all conversions, the number of significant digits must be such that accuracy is neither sacrificed nor exaggerated.

B-3.2 Precision.

First determine the intended precision. This should not be smaller than the measurement capability. (For example, a room temperature of 70°F converts to 21.111°C; however, the average thermometer could be read only with difficulty to 21.1°C, and in most cases, 21°C is the better conversion.) After estimating the intended precision, the converted value should be rounded to a minimum number of significant digits, so that a unit of the last place is equal to or smaller than the converted precision.

For example, convert 2.47-inch to metric.

$$2.47 \times 25.4 = 62.738\text{-mm}$$

Assume a precision of ± 0.005 -inch is desired. (Less than 2.465 would be rounded to 2.46; 2.475 or more would be rounded to 2.48).

$$0.005 \times 25.4 = 0.127\text{-mm}$$

A unit of 0.1-mm is smaller than 0.127-mm; therefore the conversion to intended precision is 62.7-mm (nearest tenth of mm).

However, converting 2.47-inch to cm is as follows:

$$2.47 \times 2.54 = 6.274\text{-cm}$$

Intended precision is ± 0.005 -inch.

$$0.005 \times 2.54 = 0.0127$$

A unit of 0.01 is smaller than 0.0127; therefore, the conversion to intended precision is now 6.27-cm.

B-3.3 Decimal.

It is the intended precision, not the number of decimal places in the original measurement, that determines the decimal places in the conversion. Expressed in meters, 2.47-inch is 0.0627-m. However, 2.470-inch assumes an intended precision of ± 0.0005 -inch, and converts to 62.74-mm, 6.274-cm, and 0.06274-m respectively.

B-4 NUMBER GROUPING.

B-4.1 Digits.

To make it easier to read numbers having four or more digits, the digits should be placed in groups of three separated by spaces instead of commas. Use separate groups of three both to the left and to the right of the decimal point. In case of four digits, the spacing is optional.

B-4.2 Grouping.

This style of number grouping instead of using commas, also avoids confusion caused by the use of commas to express decimal points in some countries. The conversion factors and numbers in this section are written in this style.

Examples:

1 532 or 1532	instead of	1,532
2 126 750	instead of	2,126,750
6.894 757	instead of	6.894757
0.453 592 4	instead of	0.4535924

B-5 CONVERSION USING FACTORS.

Figure B-2 has a list of general conversion factors to convert from U.S. Measure to IS metric. Multiply the given measure by the factor in the center column to get the conversion equivalent. The example in paragraph B-3.2 was done in this manner.

B-6 CONVERSION USING TABLES.

B-6.1 Tables.

Tables are provided for frequently used conversions of length, fractions, temperature, weight, torque, and pressure. The Tables for length, weight, torque, and pressure are read as shown below. The Table of fractions is self-explanatory and the use of the table of temperature conversions is explained in paragraph B-6.4.

B-6.2 Length.

Figure B-3 has the data for converting inches to millimeters.

For example, convert 2.47-inch:

From the fourth block: 2.00-inch = 50.80-mm

From the third block: 0.47-inch = 11.938-mm

$$2.47\text{-inch} = 62.738\text{-mm}$$

Rounding off to the intended accuracy, the result is 62.7-mm. (Refer to paragraph B-3.)

B-6.3 Fractions.

Figure B-4 has data for converting common fractions of an inch, or their decimal equivalents, to millimeters.

B-6.4 Temperature.

Figure B-5 has data for converting Fahrenheit and Celsius (centigrade) temperature. Find the known temperature in the column labeled C. OR F. If converting from degrees Fahrenheit to degrees Celsius, the equivalent temperature will be found in the left column. If converting from degrees Celsius to degrees Fahrenheit, the equivalent temperature will be found in the right column. A small chart of INTERPOLATION FACTORS located at the end of the table lets you convert readings within the 10° steps used above 100.

For example, convert 227°F to Celsius:

From Table: 220°F = 104°C

From Interpolation Factors Chart: 7°F = 3.89°C

$$227^{\circ}\text{F} = 108^{\circ}\text{C (rounded to intended accuracy)}$$

B-6.5 Weight (Mass).

Figure B-6 has data for converting ounces and up to 1000 pounds to kilograms. Figure B-7 has data for converting 1000 to 20000 pounds to kilograms.

IN.-LB	0	1	2	3	4	5	6
	N · m	N · m	N · m	N · m	N · m	N · m	N · m
		0.11	0.23	0.34	0.45	0.56	0.68
10	1.13	1.24	1.36	1.47	1.58	1.69	1.81
20	2.26	2.37	2.49	2.60	2.71	2.82	2.94
30	3.39	3.50	3.62	3.73	3.84	3.95	4.07
40	4.52	4.63	4.75	4.86	4.97	5.08	5.20
50	5.65	5.76	5.88	5.99	6.10	6.21	6.33
60	6.78	6.89	7.01	7.12	7.23	7.34	7.46
70	7.91	8.02	8.13	8.25	8.36	8.47	8.59
80	9.04	9.15					

(1 IN.-LB = 0.112 984 8 N · m)

USE CONVERSION CHARTS LIKE THIS:

25 INCH-POUNDS = ? N · m
 READ ACROSS THE 20 LINE TO THE 5 COLUMN AND READ
 2.82 N · m

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FIGURE B-1. TORQUE: INCH-POUNDS TO NEWTON-METERS

B-6.6 Torque.

Figure B-8 has data for converting inch-pounds to newton-meters, and Figure B-9 has data for converting foot-pounds to newton-meters.

NOTE

A pascal is equal to one newton-meter²; a kilopascal is 1000 pascals.

B-6.7 Pressure.

Figure B-10 has data for converting 1 to 100 pounds per-square-inch (psi) to kilopascals (kPa). Figure B-11 has conversion data for 100 to 10000 psi. To convert inches of water pressure, use the conversion factor given in Figure B-1.

<u>MULTIPLY</u>	<u>BY</u>	<u>TO GET</u>
<u>LENGTH</u>		
INCH	25.400 0	MILLIMETER (mm)
FOOT	0.304 8	METER (m)
YARD	0.914 4	METER (m)
MILE (STATUTE)	1.609 3	KILOMETER (km)
MILE (NAUTICAL)	1.852 0	KILOMETER (km)
<u>AREA</u>		
INCH ²	6.451 6	CENTIMETER ² (cm ²)
FOOT ²	0.092 9	METER ² (m ²)
YARD ²	0.836 1	METER ² (m ²)
<u>VOLUME</u>		
INCH ³	16.387 1	CENTIMETER ³ (cm ³)
FOOT ³	28.316 8	LITER (L)
	0.028 3	METER ³ (m ³)
OUNCE (U.S. FLUID)	29.573 5	CENTIMETER ³ (cm ³)
OUNCE (IMPERIAL)	28.413 0	CENTIMETER ³ (cm ³)
QUART (U.S. FLUID)	0.946 4	LITER (L)
GALLON (U.S. FLUID)	3.785 4	LITER (L)
GALLON (IMPERIAL)	4.546 1	LITER (L)
<u>MASS (WEIGHT)</u>		
OUNCE	28.349 5	GRAMS (g)
POUND	0.453 6	KILOGRAM (kg)
TON	907.184 7	KILOGRAM (kg)
<u>FORCE</u>		
OUNCE	0.278 0	NEWTON (N)
POUND	4.448 2	NEWTON (N)
KILOGRAM	9.806 6	NEWTON (N)
<u>TORQUE</u>		
INCH-POUND	0.113 0	NEWTON-METER (N · m)
FOOT-POUND	1.355 8	NEWTON-METER (N · m)
<u>POWER</u>		
HORSEPOWER	0.745 7	KILOWATT (kW)

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FIGURE B-2. CONVERSION FACTORS - U.S. MEASURE TO METRIC (SHEET 1 OF 2)

<u>MULTIPLY</u>	<u>BY</u>	<u>TO GET</u>
<u>PRESSURE OR STRESS</u>		
INCHES OF WATER	0.249 1	KILOPASCAL (kPa)
INCHES OF MERCURY	3.376 8	KILOPASCAL (kPa)
POUNDS PER INCH ² (PSI)	6.894 8	KILOPASCAL (kPa)
<u>ENERGY OR WORK</u>		
BTU	1055.	JOULE (J)
FOOT-POUND (FORCE)	1.355 8	JOULE (J) (J = N·m)
WATT-SECOND	1.000 0	JOULE (J)
KILOWATT-HOUR	3.6 X 10 ⁶	JOULE (J)
<u>ACCELERATION</u>		
FREE FALL	9.806 6	METER / SEC ² (m / s ²)
FOOT / SECOND ²	0.304 8	METER / SEC ² (m / s ²)
INCH / SECOND ²	0.025 4	METER / SEC ² (m / s ²)
<u>VELOCITY</u>		
KNOT	0.5144	METER / SEC (m / s)
	1.852 0	KILOMETER / HOUR (km / h)
MILE / HOUR	1.609 3	KILOMETER / HOUR (km / h)
<u>FUEL PERFORMANCE</u>		
MILE / GALLON	0.425 1	KILOMETER / LITER (km / L)
GALLON / MILE	2.352 7	LITER / KILOMETER (L / km)
<u>FLOW (MASS / TIME)</u>		
POUNDS PER HOUR	0.453 6	KILOGRAM / HOUR (Kg / h)
<u>FLOW (VOLUME / TIME)</u>		
GALLONS PER HOUR	3.785 4	LITER / HOUR (L / h)
	0.003 8	METER / HOUR (m ³ / h)
<u>LIGHT</u>		
FOOT-CANDLE	1.076 4	LUMEN / METER ² (Lm / m ²)
<u>TEMPERATURE</u>		
FAHRENHEIT	(°F - 32) X 5 / 9	= CELSIUS (°C)
CELSIUS	(°C X 9 / 5) + 32	= FAHRENHEIT (°F)

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FIGURE B-2. CONVERSION FACTORS - U.S. MEASURE TO METRIC (SHEET 2 OF 2)

INCHES	0.0000	0.0001	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009
MILLIMETERS										
0.000		0.0025	0.0050	0.0076	0.0101	0.0127	0.0152	0.0177	0.0203	0.0228
0.001	0.0254	0.0279	0.0304	0.0330	0.0355	0.0381	0.0406	0.0431	0.0457	0.0482
0.002	0.0508	0.0533	0.0558	0.0584	0.0609	0.0635	0.0660	0.0685	0.0711	0.0736
0.003	0.0762	0.0787	0.0812	0.0838	0.0863	0.0889	0.0914	0.0939	0.0965	0.0990
0.004	0.1016	0.1041	0.1066	0.1092	0.1117	0.1143	0.1168	0.1193	0.1219	0.1244
0.005	0.1270	0.1295	0.1320	0.1346	0.1371	0.1397	0.1422	0.1447	0.1473	0.1498
0.006	0.1524	0.1549	0.1574	0.1600	0.1625	0.1651	0.1676	0.1701	0.1727	0.1752
0.007	0.1778	0.1803	0.1828	0.1854	0.1879	0.1905	0.1930	0.1955	0.1981	0.2006
0.008	0.2032	0.2057	0.2082	0.2108	0.2133	0.2159	0.2184	0.2209	0.2235	0.2260
0.009	0.2286	0.2311	0.2336	0.2362	0.2387	0.2413	0.2438	0.2463	0.2489	0.2514
INCHES	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
MILLIMETERS										
0.00		0.025	0.050	0.076	0.101	0.127	0.152	0.177	0.203	0.228
0.01	0.254	0.279	0.304	0.330	0.355	0.381	0.406	0.431	0.457	0.482
0.02	0.508	0.533	0.558	0.584	0.609	0.635	0.660	0.685	0.711	0.736
0.03	0.762	0.787	0.812	0.838	0.863	0.889	0.914	0.939	0.965	0.990
0.04	1.016	1.041	1.066	1.092	1.117	1.143	1.168	1.193	1.219	1.244
0.05	1.270	1.295	1.320	1.346	1.371	1.397	1.422	1.447	1.473	1.498
0.06	1.524	1.549	1.574	1.600	1.625	1.651	1.676	1.701	1.727	1.752
0.07	1.778	1.803	1.828	1.854	1.879	1.905	1.930	1.955	1.981	2.006
0.08	2.032	2.057	2.082	2.108	2.133	2.159	2.184	2.209	2.235	2.260
0.09	2.286	2.311	2.336	2.362	2.387	2.413	2.438	2.463	2.489	2.514
INCHES	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
MILLIMETERS										
0.0		0.254	0.508	0.762	1.016	1.270	1.524	1.778	2.032	2.286
0.1	2.540	2.794	3.048	3.302	3.556	3.810	4.064	4.318	4.572	4.826
0.2	5.080	5.334	5.588	5.842	6.096	6.350	6.604	6.858	7.112	7.366
0.3	7.620	7.874	8.128	8.382	8.636	8.890	9.144	9.398	9.652	9.906
0.4	10.160	10.414	10.668	10.922	11.176	11.430	11.684	11.938	12.192	12.446
0.5	12.700	12.954	13.208	13.462	13.716	13.970	14.224	14.478	14.732	14.986
0.6	15.240	15.494	15.748	16.002	16.256	16.510	16.764	17.018	17.272	17.526
0.7	17.780	18.034	18.288	18.542	18.796	19.050	19.304	19.558	19.812	20.066
0.8	20.320	20.574	20.828	21.082	21.336	21.590	21.844	22.098	22.352	22.606
0.9	22.860	23.114	23.368	23.622	23.876	24.130	24.384	24.638	24.892	25.146
INCHES	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MILLIMETERS										
0.0		2.54	5.08	7.62	10.16	12.70	15.24	17.78	20.32	22.86
1.0	25.40	27.94	30.48	33.02	35.56	38.10	40.64	43.18	45.72	48.26
2.0	50.80	53.34	55.88	58.42	60.96	63.50	66.04	68.58	71.12	73.66
3.0	76.20	78.74	81.28	83.82	86.36	88.90	91.44	93.98	96.52	99.06
4.0	101.60	104.14	106.68	109.22	111.76	114.30	116.84	119.38	121.92	124.46
5.0	127.00	129.54	132.08	134.62	137.16	139.70	142.24	144.78	147.32	149.86
6.0	152.40	154.94	157.48	160.02	162.56	165.10	167.64	170.18	172.72	175.26
7.0	177.80	180.34	182.88	185.42	187.96	190.50	193.04	195.58	198.12	200.66
8.0	203.20	205.74	208.28	210.82	213.36	215.90	218.44	220.98	223.52	226.06
9.0	228.60	231.14	233.68	236.22	238.76	241.30	243.84	246.38	248.92	251.46

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FIGURE B-3. INCHES TO MILLIMETERS (0.0001-INCH TO 10-INCHES)

INCH	1/2's	1/4's	8ths	16ths	32nds	64ths	MILLIMETERS	DECIMALS OF AN INCH*
						1	0.397	0.015 625
					1	2	0.794	0.031 25
						3	1.191	0.046 875
				1	2	4	1.588	0.062 5
						5	1.984	0.078 125
					3	6	2.381	0.093 75
						7	2.778	0.109 375
			1	2	4	8	3.175*	0.125 0
						9	3.572	0.140 625
					5	10	3.969	0.156 25
						11	4.366	0.171 875
				3	6	12	4.762	0.187 5
						13	5.159	0.203 125
					7	14	5.556	0.218 75
						15	5.953	0.234 375
		1	2	4	8	16	6.350*	0.250 0
						17	6.747	0.265 625
						9	7.144	0.281 25
						19	7.541	0.296 875
				5	10	20	7.938	0.312 5
						21	8.334	0.328 125
					11	22	8.731	0.343 75
						23	9.128	0.359 375
			3	6	12	24	9.525*	0.375 0
						25	9.922	0.390 625
					13	26	10.319	0.406 25
						27	10.716	0.421 875
				7	14	28	11.112	0.437 5
						29	11.509	0.453 125
					15	30	11.906	0.468 75
						31	12.303	0.484 375
	1	2	4	8	16	32	12.700*	0.500 0
*EXACT								

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FIGURE B-4. COMMON FRACTIONS AND DECIMALS TO MILLIMETERS (SHEET 1 OF 2)

INCH	1/2's	1/4's	8ths	16ths	32nds	64ths	MILLIMETERS	DECIMALS OF AN INCH*
						33	13.097	0.515 625
					17	34	13.494	0.531 25
						35	13.891	0.546 875
				9	18	36	14.288	0.562 5
						37	14.984	0.578 125
					19	38	15.081	0.593 75
						39	15.478	0.609 375
			5	10	20	40	15.875*	0.625 0
						41	16.272	0.640 625
					21	42	16.669	0.656 25
						43	17.066	0.671 875
				11	22	44	17.462	0.687 5
						45	17.859	0.703 125
					23	46	18.256	0.718 75
						47	18.653	0.734 375
		3	6	12	24	48	19.050*	0.750 0
						49	19.447	0.765 625
					25	50	19.844	0.781 25
						51	20.241	0.796 875
				13	26	52	20.638	0.812 5
						53	21.034	0.828 125
					27	54	21.431	0.843 75
						55	21.828	0.859 375
			7	14	28	56	22.225*	0.875 0
						57	22.622	0.890 625
					29	58	23.019	0.906 25
						59	23.416	0.921 875
				15	30	60	23.812	0.937 5
						61	24.209	0.953 125
					31	62	24.606	0.968 75
						63	25.003	0.984 375
1	2	4	8	16	32	64	25.400*	1.000 0
*EXACT								

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FIGURE B-4. COMMON FRACTIONS AND DECIMALS TO MILLIMETERS (SHEET 2 OF 2)

-459.4 TO -220			-210 TO 0			1 TO 25			26 TO 50		
C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.
-273	-459.4		-134	-210	-346	-17.2	1	33.8	-3.33	26	78.8
-268	-450		-129	-200	-328	-16.7	2	35.6	-2.78	27	80.6
-262	-440		-123	-190	-310	-16.1	3	37.4	-2.22	28	82.4
-257	-430		-118	-180	-292	-15.6	4	39.2	-1.67	29	84.2
-251	-420		-112	-170	-274	-15.0	5	41.0	-1.11	30	86.0
-246	-410		-107	-160	-256	-14.4	6	42.8	-0.56	31	87.8
-240	-400		-101	-150	-238	-13.9	7	44.6	0	32	89.6
-234	-390		-95.6	-140	-220	-13.3	8	46.4	0.56	33	91.4
-229	-380		-90.0	-130	-202	-12.8	9	48.2	1.11	34	93.2
-223	-370		-84.4	-120	-184	-12.2	10	50.0	1.67	35	95.0
-218	-360		-78.9	-110	-166	-11.7	11	51.8	2.22	36	96.8
-212	-350		-73.3	-100	-148	-11.1	12	53.6	2.78	37	98.6
-207	-340		-67.8	-90	-130	-10.6	13	55.4	3.33	38	100.4
-201	-330		-62.2	-80	-112	-10.0	14	57.2	3.89	39	102.2
-196	-320		-56.7	-70	-94	-9.44	15	59.0	4.44	40	104.0
-190	-310		-51.1	-60	-76	-8.89	16	60.8	5.00	41	105.8
-184	-300		-45.6	-50	-58	-8.33	17	62.6	5.56	42	107.6
-179	-290		-40.0	-40	-40	-7.78	18	64.4	6.11	43	109.4
-173	-280		-34.4	-30	-22	-7.22	19	66.2	6.67	44	111.2
-169	-273	-459.4	-28.9	-20	-4	-6.67	20	68.0	7.22	45	113.0
-168	-270	-454	-23.3	-10	14	-6.11	21	69.8	7.78	46	114.8
-163	-260	-436	-17.8	0	32	-5.56	22	71.6	8.33	47	116.6
-157	-250	-418				-5.00	23	73.4	8.89	48	118.4
-151	-240	-400				-4.44	24	75.2	9.44	49	120.2
-146	-230	-382				-3.89	25	77.0	10.0	50	122.0
-140	-220	-364									

51 TO 75			76 TO 100			101 TO 340			341 TO 490			491 TO 750		
C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.
10.6	51	123.8	24.4	76	168.8	43	110	230	177	350	662	260	500	932
11.1	52	125.6	25.0	77	170.6	49	120	248	182	360	680	266	510	950
11.7	53	127.4	25.6	78	172.4	54	130	266	188	370	698	271	520	968
12.2	54	129.2	26.1	79	174.2	60	140	284	193	380	716	277	530	986
12.8	55	131.0	26.7	80	176.0	66	150	302	199	390	734	282	540	1004
13.3	56	132.8	27.2	81	177.8	71	160	320	204	400	752	288	550	1022
13.9	57	134.6	27.8	82	179.6	77	170	338	210	410	770	293	560	1040
14.4	58	136.4	28.3	83	181.4	82	180	356	216	420	788	299	570	1058
15.0	59	138.2	28.9	84	183.2	88	190	374	221	430	806	304	580	1076
15.6	60	140.0	29.4	85	185.0	93	200	392	227	440	824	310	590	1094
16.1	61	141.8	30.0	86	186.8	99	210	410	232	450	842	316	600	1112
16.7	62	143.6	30.6	87	188.6	100	212	413	238	460	860	321	610	1130
17.2	63	145.4	31.1	88	190.4	104	220	428	243	470	878	327	620	1148
17.8	64	147.2	31.7	89	192.2	110	230	446	249	480	896	332	630	1166
18.3	65	149.0	32.2	90	194.0	116	240	464	254	490	914	338	640	1184
18.9	66	150.8	32.8	91	195.8	121	250	482				343	650	1202
19.4	67	152.6	33.3	92	197.6	127	260	500				349	660	1220
20.0	68	154.4	33.9	93	199.4	132	270	518				354	670	1238
20.6	69	156.2	34.4	94	201.2	138	290	536				360	680	1256
21.1	70	158.0	35.0	95	203.0	143	290	554				366	690	1274
21.7	71	159.8	35.6	96	204.8	149	300	572				371	700	1292
22.2	72	161.6	36.1	97	206.6	154	310	590				377	710	1310
22.8	73	163.4	36.7	98	208.4	160	320	608				382	720	1328
23.3	74	165.2	37.2	99	210.2	166	330	626				388	730	1346
23.9	75	167.0	37.8	100	212.0	171	340	644				393	740	1364
												399	750	1382

FB1631_1
SAFIGURE B-5. FAHRENHEIT AND CELSIUS (CENTIGRADE) TEMPERATURE CONVERSION
(SHEET 1 OF 3)

751 TO 1000			1001 TO 1250			1251 TO 1490			1491 TO 1750		
C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.
404	760	1400	543	1010	1850	682	1260	2300	816	1500	2732
410	770	1418	549	1020	1868	688	1270	2318	821	1510	2750
416	780	1436	554	1030	1886	693	1280	2336	827	1520	2768
421	790	1454	560	1040	1904	699	1290	2354	832	1530	2786
427	800	1472	566	1050	1922	704	1300	2372	838	1540	2804
432	810	1490	571	1060	1940	710	1310	2390	843	1550	2822
438	820	1508	577	1070	1958	716	1320	2408	849	1560	2840
443	830	1526	582	1080	1976	721	1330	2426	854	1570	2858
449	840	1544	588	1090	1994	727	1340	2444	860	1580	2876
454	850	1562	593	1100	2012	732	1350	2462	866	1590	2894
460	860	1580	599	1110	2030	738	1360	2480	871	1600	2912
466	870	1598	604	1120	2048	743	1370	2498	877	1610	2930
471	880	1616	610	1130	2066	749	1380	2516	882	1620	2948
477	890	1634	616	1140	2084	754	1390	2534	888	1630	2966
482	900	1652	621	1150	2102	760	1400	2552	893	1640	2984
488	910	1670	627	1160	2120	766	1410	2570	899	1650	3002
493	920	1688	632	1170	2138	771	1420	2588	904	1660	3020
499	930	1706	638	1180	2156	777	1430	2606	910	1670	3038
504	940	1724	643	1190	2174	782	1440	2624	916	1680	3056
510	950	1742	649	1200	2192	788	1450	2642	921	1690	3074
516	960	1760	654	1210	2210	793	1460	2660	927	1700	3092
521	970	1778	660	1220	2228	799	1470	2678	932	1710	3110
527	980	1796	666	1230	2246	804	1480	2696	938	1720	3128
532	990	1814	671	1240	2264	810	1490	2714	943	1730	3146
538	1000	1832	677	1250	2282				949	1740	3164
									954	1750	3182

FB1631_2
SA

**FIGURE B-5. FAHRENHEIT AND CELSIUS (CENTIGRADE) TEMPERATURE CONVERSION
(SHEET 2 OF 3)**

1751 TO 2000			2001 TO 2250			2251 TO 2490			2491 TO 2750			2751 TO 3000		
C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.	C.	C. OR F.	F.
960	1760	3200	1099	2010	3650	1238	2260	4100	1371	2500	4532	1516	2760	5000
966	1770	3218	1104	2020	3668	1243	2270	4118	1377	2510	4550	1521	2770	5018
971	1780	3236	1110	2030	3686	1249	2280	4136	1382	2520	4568	1527	2780	5036
977	1790	3254	1116	2040	3704	1254	2290	4154	1388	2530	4586	1532	2790	5054
982	1800	3272	1121	2050	3722	1260	2300	4172	1393	2540	4604	1538	2800	5072
988	1810	3290	1127	2060	3740	1266	2310	4190	1399	2550	4622	1543	2810	5090
993	8120	3308	1132	2070	3758	1271	2320	4208	1404	2560	4640	1549	2820	5184
999	1830	3326	1138	2080	3776	1277	2330	4226	1410	2570	4658	1554	2830	5126
1004	1840	3344	1143	2090	3794	1282	2340	4244	1416	2580	4676	1560	2840	5144
1010	1850	3362	1149	2100	3812	1288	2350	4262	1421	2590	4694	1566	2850	5162
1016	1860	3380	1154	2110	3830	1293	2360	4280	1427	2600	4712	1571	2860	5180
1021	1870	3398	1160	2120	3848	1299	2370	4298	1432	2610	4730	1577	2870	5198
1027	1880	3416	1166	2130	3866	1304	2380	4316	1438	2620	4748	1582	2880	5216
1032	1890	3434	1171	2140	3884	1310	2390	4334	1443	2630	4766	1588	2890	5234
1038	1900	3452	1177	2150	3902	1316	2400	4352	1449	2640	4784	1593	2900	5252
1043	1910	3470	1182	2160	3920	1321	2410	4370	1454	2650	4802	1599	2910	5270
1049	1920	3488	1188	2170	3938	1327	2420	4388	1460	2660	4820	1604	2920	5288
1054	1930	3506	1193	2180	3956	1332	2430	4406	1466	2670	4838	1610	2930	5306
1060	1940	3524	1199	2190	3974	1338	2440	4424	1471	2680	4856	1616	2940	5324
1066	1950	3542	1204	2200	3992	1343	2450	4442	1477	2690	4874	1621	2950	5342
1071	1960	3560	1210	2210	4010	1349	2460	4460	1482	2700	4892	1627	2960	5360
1077	1970	3578	1216	2220	4028	1354	2470	4478	1488	2710	4910	1632	2970	5378
1082	1980	3596	1221	2230	4046	1360	2480	4496	1493	2720	4928	1638	2980	5396
1088	1990	3614	1227	2240	4064	1366	2490	4514	1499	2730	4946	1643	2990	5414
1093	2000	3632	1232	2250	4082				1504	2740	4965	1649	3000	5432
									1510	2750	4982			

NOTE

THE NUMBERS IN COLUMN LABELED "C. OR F." REFER TO THE TEMPERATURE EITHER IN DEGREES CELSIUS OR FAHRENHEIT WHICH IT IS DESIRED TO CONVERT INTO THE OTHER SCALE. IF CONVERTING FROM DEGREES FAHRENHEIT TO DEGREES CELSIUS THE EQUIVALENT TEMPERATURE WILL BE FOUND IN THE LEFT COLUMN, WHILE IF CONVERTING FROM DEGREES CELSIUS TO DEGREES FAHRENHEIT, THE ANSWER WILL BE FOUND IN THE COLUMN ON THE RIGHT.

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5 / 9$$

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 9 / 5) + 32$$

INTERPOLATION FACTORS	C.		F.		C.		F.	
	0.56	1	1.8		3.33	6	10.8	
	1.11	2	3.6		3.89	7	12.6	
	1.67	3	5.4		4.44	8	14.4	
	2.22	4	7.2		5.00	9	16.2	
	2.78	5	9.0		5.56	10	18.0	

FB1631_3C
SA

**FIGURE B-5. FAHRENHEIT AND CELSIUS (CENTIGRADE) TEMPERATURE CONVERSION
(SHEET 3 OF 3)**

	0	1	2	3	4	5	6	7	8	9
	kg	kg	kg	kg	kg	kg	kg	kg	kg	kg
OZ										
0	–	0.028	0.057	0.085	0.113	0.142	0.170	0.198	0.227	0.255
10	0.283	0.312	0.340	0.369	0.397	0.425	0.454	0.482	0.510	0.539
LB										
0	–	0.45	0.91	1.36	1.81	2.27	2.72	3.18	3.63	4.08
10	4.5	5.0	5.4	5.9	6.4	6.8	7.3	7.7	8.2	8.6
20	9.1	9.5	10.0	10.4	10.9	11.3	11.8	12.2	12.7	13.2
30	13.6	14.1	14.5	15.0	15.4	15.9	16.3	16.8	17.2	17.7
40	18.1	18.6	19.1	19.5	20.0	20.4	20.9	21.3	21.8	22.2
50	22.7	23.1	23.6	24.0	24.5	24.9	25.4	25.9	26.3	26.8
60	27.2	27.7	28.1	28.6	29.0	29.5	29.9	30.4	30.8	31.3
70	31.8	32.2	32.7	33.1	33.6	34.0	34.5	34.9	35.4	35.8
80	36.3	36.7	37.2	37.6	38.1	38.6	39.0	39.5	39.9	40.4
90	40.8	41.3	41.7	42.2	42.6	43.1	43.5	44.0	44.5	44.9
100	45	46	46	47	47	48	48	49	49	49
	0	10	20	30	40	50	60	70	80	90
200	91	95	100	104	109	113	118	122	127	132
300	136	141	145	150	154	159	163	168	172	177
400	181	186	191	195	200	204	209	213	218	222
500	227	231	236	240	245	249	254	259	263	268
600	272	277	281	286	290	295	299	304	308	313
700	318	322	327	331	336	340	345	349	354	358
800	363	367	372	376	381	386	390	395	399	404
900	408	413	417	422	426	431	435	440	445	449
1000	454	458	463	467	472	476	481	485	490	494

(1 OZ = 0.028 349 52 kg) (1 LB = 0.453 592 4 kg)

FB1632
SA

FIGURE B-6. WEIGHT (MASS): OUNCES OR POUNDS TO KILOGRAMS

LB	0	100	200	300	400	500	600	700	800	900
(000)*	kg	kg	kg	kg	kg	kg	kg	kg	kg	kg
1	454	499	544	590	635	680	726	771	816	862
2	907	953	998	1043	1089	1134	1179	1225	1270	1315
3	1361	1406	1451	1497	1542	1588	1633	1678	1724	1769
4	1814	1860	1905	1950	1996	2041	2087	2132	2177	2223
5	2268	2313	2359	2404	2449	2495	2540	2585	2631	2676
6	2722	2767	2812	2858	2903	2948	2994	3039	3084	3130
7	3175	3221	3266	3311	3357	3402	3447	3493	3538	3583
8	3629	3674	3719	3765	3810	3856	3901	3946	3992	4037
9	4082	4128	4173	4218	4264	4309	4354	4400	4445	4491
10	4536	4581	4627	4672	4717	4763	4803	4853	4899	4944
11	4990	5035	5080	5126	5171	5216	5262	5307	5352	5398
12	5443	5488	5534	5579	5625	5670	5715	5761	5806	5851
13	5897	5942	5987	6033	6078	6123	6169	6214	6260	6305
14	6350	6369	6441	6486	6532	6577	6622	6668	6713	6759
15	6804	6849	6895	6940	6985	7031	7076	7121	7167	7212
16	7257	7303	7348	7394	7439	7484	7530	7575	7620	7666
17	7711	7756	7802	7847	7893	7938	7983	8029	8074	8119
18	8165	8210	8255	8301	8346	8391	8437	8482	8528	8573
19	8618	8664	8709	8754	8800	8845	8890	8936	8981	9026
20	9072	9117	9163	9208	9253	9299	9344	9389	9435	9480
*MULTIPLY LB VALUE BY 1000										

(1 LB = 0.453 592 4 kg)

FB1633
SA

FIGURE B-7. WEIGHT (MASS): THOUSAND POUNDS TO KILOGRAMS

IN.-LB	0	1	2	3	4	5	6	7	8	9
	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m
10	1.13	0.11	0.23	0.34	0.45	0.56	0.68	0.79	0.90	1.02
20	2.26	1.24	1.36	1.47	1.58	1.69	1.81	1.92	2.03	2.15
30	3.39	2.37	2.49	2.60	2.71	2.82	2.94	3.05	3.16	3.28
40	4.52	3.50	3.62	3.73	3.84	3.95	4.07	4.18	4.29	4.41
50	5.65	4.63	4.75	4.86	4.97	5.08	5.20	5.31	5.42	5.54
60	6.78	5.76	5.88	5.99	6.10	6.21	6.33	6.44	6.55	6.67
70	7.91	6.89	7.01	7.12	7.23	7.34	7.46	7.57	7.68	7.80
80	9.04	8.02	8.13	8.25	8.36	8.47	8.59	8.70	8.81	8.93
90	10.17	9.15	9.26	9.38	9.49	9.60	9.72	9.83	9.94	10.06
100	11.30	10.28	10.39	10.51	10.62	10.73	10.85	10.96	11.07	11.19
	11.30	11.41	11.52	11.64	11.75	11.86	11.98	12.09	12.20	12.32
	0	10	20	30	40	50	60	70	80	90
200	22.60	23.73	24.86	25.99	27.12	28.25	29.38	30.51	31.64	32.77
300	33.90	35.03	36.16	37.28	38.41	39.54	40.67	41.80	42.93	44.06
400	45.19	46.32	47.45	48.58	49.71	50.84	51.97	53.10	54.23	55.36
500	56.49	57.62	58.75	59.88	61.01	62.14	63.27	64.40	65.53	66.66

(1 IN.-LB = 0.112 984 8 N · m)

FB1634
SA

FIGURE B-8. TORQUE: INCH-POUNDS TO NEWTON-METERS

FT-LB	0	1	2	3	4	5	6	7	8	9
	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m	N · m
10	14	15	16	18	19	20	22	23	24	26
20	27	28	30	31	33	34	35	37	38	39
30	41	42	43	45	46	47	49	50	52	53
40	54	56	57	58	60	61	62	64	65	66
50	68	69	71	72	73	75	76	77	79	80
60	81	83	84	85	87	88	89	91	92	94
70	95	96	98	99	100	102	103	104	106	107
80	108	110	111	113	114	115	117	118	119	121
90	122	123	125	126	127	129	130	132	133	134
FT-LB		100	200	300	400	500	600	700	800	900
N · m		136	271	407	542	678	813	949	1085	1220
FT-LB		1000	2000	3000	4000	5000	6000	7000	8000	9000
N · m		1356	2712	4067	5423	6779	8135	9491	10 847	12 202

(1 FT-LB = 1.355 838 N · m)

FB1635
SA

FIGURE B-9. TORQUE: FOOT-POUNDS TO NEWTON-METERS

PSI	0	100	200	300	400	500	600	700	800	900
(000)*	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa
		689	1 379	2 068	2 758	3 447	4 137	4 826	5 516	6 205
1	6 895	7 584	8 274	8 963	9 653	10 342	11 032	11 721	12 411	13 100
2	13 790	14 479	15 168	15 858	16 547	17 237	17 926	18 616	19 305	19 995
3	20 684	21 374	22 063	22 753	23 442	24 132	24 821	25 511	26 200	26 890
4	27 579	28 268	28 958	29 647	30 337	31 026	31 716	32 405	33 095	33 784
5	34 474	35 163	35 853	36 542	37 232	37 921	38 611	39 300	39 990	40 678
6	41 369	42 058	42 747	43 437	44 126	44 816	45 505	46 195	46 884	47 574
7	48 263	48 953	49 642	50 332	51 021	51 711	52 400	53 090	53 779	54 469
8	55 158	55 848	56 537	57 226	57 916	58 605	59 295	59 984	60 674	61 363
9	62 053	62 742	63 432	64 121	64 811	65 500	66 190	66 879	67 569	68 258
10	68 948	69 637	70 327	71 016	71 705	72 395	73 084	73 774	74 463	75 153
*MULTIPLY PSI VALUE BY 1000										

(1 PSI = 6.894 757 kPa)

FB1636
SA**FIGURE B-10. PRESSURE: THOUSAND POUNDS PER-SQUARE-INCH TO KILOPASCALS**

PSI	0	1	2	3	4	5	6	7	8	9
	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa	kPa
		6.9	13.8	20.7	27.6	34.5	41.4	48.3	55.2	62.1
10	69	76	83	90	97	103	110	117	124	131
20	138	145	152	159	165	172	179	186	193	200
30	207	214	221	228	234	241	248	255	262	269
40	276	283	290	296	303	310	317	324	331	338
50	345	352	359	365	372	379	386	393	400	407
60	414	421	427	434	441	448	455	462	469	476
70	483	490	496	503	510	517	524	531	538	545
80	552	558	565	572	579	586	593	600	607	614
90	621	627	634	641	648	655	662	669	676	683
100	689	696	703	710	717	724	731	738	745	752

(1 PSI – 6.894 757 kPa)

FB2197
SA

FIGURE B-11. PRESSURE: POUNDS PER-SQUARE-INCH TO KILOPASCALS